THE ANATOMY AND PHYSIOLOGY OF ERROR IN ADVERSE HEALTH CARE EVENTS

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ABSTRACT

Recent reports by the Institute of Medicine (IOM) signal a substantial yet unrealized deficit in patient safety innovation and improvement. With the aim of reducing this dilemma, we provide an introductory account of clinical error resulting from poorly designed systems by reviewing the relevant health care, management, psychology, and organizational accident sciences literature. First, we discuss the concept of health care error and describe two approaches to analyze error proliferation and causation. Next, by applying transdisciplinary evidence and knowledge to health care, we detail the attributes fundamental to constructing safer health care systems as embedded components within the complex adaptive environment. Then, the Health Care Error Proliferation Model *explains the sequence of events typically leading to adverse outcomes, emphasizing the role that organizational and external cultures contribute to error identification, prevention, mitigation, and defense construction. Subsequently, we discuss the critical contribution health care leaders can make to address error as they strive to position their institution as a high*

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reliability organization (HRO). Finally, we conclude that the future of patient safety depends on health care leaders adopting a system philosophy of error management, investigation, mitigation, and prevention. This change is accomplished when leaders apply the basic organizational accident and health care safety principles within their respective organizations.

INTRODUCTION

The Institute of Medicine (IOM) established the contemporary starting line for the national patient safety movement with the seminal report, *To Err is Human: Building a Safer Health System*. The IOM estimated 98,000 patients die annually in American hospitals as a result of medical errors (Kohn, Corrigan, & Donaldson, 2000). Subsequent research using additional resources indicated that the number of preventable deaths was closer to 200,000 each year (Zhan & Miller, 2003). Based on their analysis, the IOM depicted the overall state of health care as a system that frequently harms and routinely fails to deliver the appropriate standard of care (Davis et al., 2002).

In the years following the IOM report, tremendous public and political pressures motivated health care organizations to identify and attempt to reduce adverse events (Berta & Baker, 2004; Wachter, 2004). Indeed, patient safety advocates point to the IOM report as "galvanizing a dramatically expanded level of conversation and concern about patient injuries in health care both in the United States and abroad" (Leape & Berwick, 2005, p. 2384). Patient safety arrived at the point of being "a national problem that became increasingly difficult for providers to ignore" (Devers, Pham, & Lui, 2004, p. 103). Regulatory (e.g. CMS), accreditation (e.g. Joint Commission), and quality-improvement organizations (e.g. NCQA) develop and advocate patient safety standards primarily derived from expert panel recommendations and opinions with the majority lacking evidentiary knowledge (Agency for Healthcare Research and Quality, 2001; Institute of Medicine, 2001).

Although health care organizations acknowledged the importance of *To Err is Human*, the majority of leaders "expressed different levels of commitment to patient safety" (Devers et al., 2004, p. 111) reflected by increasing patient morbidity and mortality related to adverse events (Institute of Medicine, 2004; Zhan & Miller, 2003). Devers and colleagues

characterized the impact of the patient safety movement on improving health care systems as "occurring relatively slowly and incrementally" (2004, p. 114). By most accounts, the patient safety movement requires transformation in the areas of error identification, process improvement, and cultural renovation to defend patients from harm (Committee on the Future of Rural Healthcare, 2005; Institute of Medicine, 2001, 2003, 2004, 2007a, 2007b; Kohn et al., 2000).

Reducing Errors

The purpose of this chapter is to describe the current state of the often unreliable health care system and to provide an introductory account of error resulting from poorly designed systems. In doing so, we review the relevant health care, management, psychology, and organizational accident science literature and synthesize an error framework from a transdisciplinary perspective. First, we discuss the concept of error and the complexities associated with errors within the complex adaptive health care system. Second, we describe two approaches to analyze error causation and discuss the associated implications. Next, we summarize the *Swiss Cheese Model* of adverse events (Reason, 1990, 2000), advocating modifications to emphasize both the impact of the complex adaptive system on health care professionals and the role error serves to distract leaders and clinicians from system improvements. Through this transdisciplinary approach, we present our *Health Care Error Proliferation Model*. This adaptation of the *Swiss Cheese Model* improves and updates the applicability of the general structural elements specifically to health care. Finally, we discuss the critical role health care leadership serves to proactively address error causation. Ultimately, the emphasis on error reduction and system defense strategies will lead to a minimal number of adverse events, an attribute associated with high reliability organizations (HROs) and industries.

PHYSIOLOGY OF ERROR

In this section, we review two approaches to dissecting the root cause of error: the person approach and the system approach. These two approaches represent distinct philosophies of error causation which lead to divergent methods for managing liability and reducing errors.

Person Approach

The person approach advocates identifying the culpable party as the root cause of an adverse event (Reason, 1998). Historically, this health care error investigation process focuses on the "who did it" instead of the "why did it happen" (Kohn et al., 2000; Rasmussen, 1999; Reason, 2000). The person approach is commonly preferred because "blaming individuals is emotionally more satisfying than targeting institutions" (Reason, 2000, p. 70). A leading rationale to explicate this person focus is fundamental attribution theory. Attribution theory is a body of knowledge about how people explain things by either internal or external factors (Heider, 1958; Weiner, 1972). Generally, attribution theory describes the process whereby leaders seek: (1) to understand the cause of an event (Heider, 1958), (2) assess responsibility for the outcomes (Martinko & Gardner, 1982), and (3) appraise the personal attributes of the involved parties (Weiner, 1995b). These three attributes allow leaders to rapidly form determinations to describe employee behavior and performance (Harvey & Weary, 1985). Attribution theory explains leaders' and managers' thinking processes as related to conclusions about the development and causation of an event (Martinko, Douglas, & Harvey, 2006; Martinko & Thomson, 1998). Essential to attribution theory is the method by which leaders form opinions about worker's performance (Martinko & Thomson, 1998; Weiner, 1995a).

Attribution is synonymous with explaining "why did something occur" (Green & Mitchell, 1979) in the context of the complex adaptive environment. As such, attributing error to a particular clinician is quick, easy, and efficient for leaders to formulate immediate conclusions. Although swift in generating answers, the accuracy of error attribution is troublesome within complex systems given the plethora of variables, processes, and functions (Dorner, 1996). In addition, speed matters while accuracy is less important following an accident because attributing error to the clinician protects the organization from needing to admit culpability during the mitigation phase of the risk management process. With further scrutiny, however, we discover the majority of errors leading to patient harm are not related to incompetent or substandard clinician care (Cook & Woods, 1994; Kohn et al., 2000; Reason, Carthey, & de Leval, 2001). Rather, errors more frequently reflect an inability of clinicians and other health care workers to cope with multiple gaps produced by system complexity (Dorner, 1996; Wiegmann & Shappell, 1999; Woods & Cook, 2002).

Aviation is an industry that has achieved a remarkably low error rate. Reports attribute 80–90% of all errors to system complexity (Wiegmann &

Shappell, 1999). Experts hypothesize that this complexity is similar for the health care industry (Helmreich, 2000; Helmreich & Davies, 2004; Kohn et al., 2000). In other words, the root causes of error are not prominently located at individual level but tend to be a system property requiring cause-and-effect analysis to elucidate "the causes" (Gano, 2003; Reason, 2000). Often, these system flaws, including obvious and dangerous conditions, remain undiscovered, hidden, or invisible until a sentinel event results. Described simply as an "error with sad consequences" (Cherns, 1962), most accidents result in the course of a complex chain reaction with a triggering cause or causes. In health care, a serious accident is called a sentinel event. Sentinel events are characterized as the unanticipated adverse event resulting in an outcome or potential outcome of serious injury or death not related to a patient's expected illness trajectory (Joint Commission Resources, 2007).

By supporting the person approach, clinical professionals, health care leaders, professional boards, and even the public frequently consider practitioners involved in errors to be "at fault" for neglecting to protect the patient. In fact, the National Council of State Boards of Nursing (NCSBN) could further exacerbate the assignment of blame by the person approach in describing the types and sources of nursing error through an inductive process called "practice breakdowns" (National Council of State Boards of Nursing, 2007). One universal consequence of adopting the person approach to error management is the culture of fear it engenders (Kohn et al., 2000; Reason, 2000; Rosenthal, 1994).

The term "practice breakdowns" is defined as "the disruption or absence of any of the aspects of good practice" (Benner et al., 2006, p. 53). By categorizing error in terms of "practice breakdowns," without specifically describing or defining what the disruption to, or the absence of, good practice means, error will continue to be adjudicated by examining performance closest to the adverse event, at the clinician practice level, as separate and removed components from the greater complex adaptive system. Perrow (1984) rejected the term "operator breakdown" in describing an accident because this blames the worker. The IOM (Kohn et al., 2000, p. 43) emphasizes the "individual provider issue" rather than failures in the process of providing care in a complex system as problematic to improving health care delivery. In short, the term "practice breakdowns" suggests a "clinician fault" connotation and may slow down the improvement to health care.

Furthermore, public blame for clinician error and the subsequent punishment for adverse events due to error are expected by the majority of health care leaders regardless of the root causes (Vincent, 1997). The result of this punitive culture is that doctors and nurses hide "practice

breakdown'' situations (Kohn et al., 2000; Lawton & Parker, 2002) and mistakes to avoid reprisal and punishment (Gibson & Singh, 2003; Kohn et al., 2000). Speaking in opposition to the person approach, Gibson and Singh (2003, p. 24) stated, ''When a health care professional reports a medical error, they suffer intimidation. They lose their standing, their status and are ostracized. An impenetrable culture of silence has developed…''

Similarly, this person approach was illustrated by a recent sentinel event at a Midwestern hospital. An experienced labor and delivery nurse mistakenly delivered an epidural drug to a young late-term pregnant patient through the intravenous line (Institute for Safe Medication Practices, 2006). Unfortunately, the patient died as a direct result of this critical sentinel event error at the sharp-end, near the bedside. However, closer analysis revealed that the fatal nursing error was probably only one in a cascade of events plagued with ''hidden'' or contributory errors. Typically observed in medication delivery systems, errors originate proximal to system level and cascade through the process distally to clinical practice (Smetzer & Cohen, 2006).

Due to the collective inability to recognize the nursing error in the context of a complex adaptive system with abundant dormant conditions, the nurse was subjected to significant public blame creating humiliation and hardship as well as criminal prosecution (Institute for Safe Medication Practices, 2006; State of Wisconsin, 2006). The nurse faced serious punishment despite the presence of other significant contributory factors (Institute for Safe Medication Practices, 2006). Numerous professional and quality-improvement organizations reacted with position statements objecting to the miscarriage of justice (Institute for Safe Medication Practices, 2006; Wisconsin Hospital Association, 2006; Wisconsin Medical Society, 2006), which is not in the spirit of *To Err is Human* (Institute for Safe Medication Practices, 2006; Kohn et al., 2000; Wisconsin Medical Society, 2006).

In fact, the continual application of blame, punishment, and shame to address health care error facilitates the cultural evolution of learned helplessness (Seligman, 1965; Garber & Seligman, 1980; Seligman, Maier, & Geer, 1968). Learned helplessness is an evolutionary process where clinicians become passive professionals as a consequence of repeated punishment for errors that make success unlikely even following organizational change (Abramson, Garber, & Seligman, 1980; Martinko & Gardner, 1982). Repeated punishment for clinical errors may be internalized (Garber & Seligman, 1980) as a result of practitioners frequently witnessing the adjudication of active error via ''blame and shame'' with chastisement and character assassination. Outside the organization walls, sensationalized media reports only aggravate this dilemma. As a result, clinicians may develop a lack

of self-confidence in their abilities to perform without punishment for mistakes resulting in deterioration of performance (Peterson, Maier, & Seligman, 1993). Fashioned by this sharp system quality, the condition may be exacerbated by quasi-legal entities, such as the NCSBN, distinguishing error using the person approach and viewing error as a professional practice characteristics as opposed to a common system attribute.

System Approach

The system approach stands in contrast to the person approach in the philosophy related to error and adverse event adjudication, investigation, and mitigation. According to Reason, the system approach to error management unearths concealed (latent) errors (2000) and exposes vulnerability through intensive system evaluation (Reason et al., 2001), while discounting the visible (active) errors caused by being human. The system approach advocates that while individual practitioners "must be responsible for the quality of their work, more errors will be eliminated by focusing on systems than on individuals" (Leape et al., 1995, p. 40). Consequently, this approach relies on investigative techniques and transdisciplinary analysis of both latent and active errors as threats to the system (Helmreich, 2000). Systems thinking and proactive process improvement on the part of health care organizations remains a significant opportunity for advancement (Amalberti, Auroy, Berwick, & Barach, 2005).

The vast majority of errors that contribute to accidents results from poorly designed systems (Cook, Render, & Woods, 2000; Rasmussen, 1990; Reason, 2000) rather than from the carelessness or neglect of professionals working within the environment (Cook & Woods, 1994; Helmreich & Davies, 2004). The IOM recognized this approach in the report, *Crossing the Quality Chasm* (2001, p. 4) stating, "Trying harder will not work. Changing systems of care will."

The responsiveness of clinicians and hospital leaders to the numerous patient safety calls have been slow to emerge (Leape & Berwick, 2005). In response to the IOM's call for comprehensive system transformation, Millenson (2003, p. 104) identified a barrier related to system improvement, stating "the IOM's focus on 'system' improvement ignores the repeated refusal by physicians and hospital leaders to adopt [better] systems" in their effort to improve patient safety. Most clinicians believe they are already working to improve the system when they are not. This is supported by Devers et al. (2004, p. 111) categorization of physicians as "barrier[s] failing to buy into the magnitude of the [safety] problem."

Deeply embedded but masked features, or latent factors (Dorner, 1996; Perrow, 1984; Reason, 2000) are the primary leadership focus subsequent to an adverse event (Reason, 1990). Although no system can completely eliminate error (Perrow, 1984; Rasmussen, 1990), the system approach is an instrumental philosophy to tackle potential hazards aimed at reducing risk, increasing reliability, and improving quality (Kohn et al., 2000; Reason, 2000; Smetzer & Cohen, 1998).

HROs are prime examples of the system approach (Reason, 2000; Roberts, 2002) put into organizational practice. These industries are considered to be high risk, such as airlines and nuclear power plants, whose system design, organizational culture, and leadership commitment to safety facilitate highly reliable, low accident, systems (Roberts, 1990; Weick & Sutcliffe, 2001). The achievement and maintenance of exceptionally low operational process variation is the key distinguishing feature among HROs (Roberts, 1990). Although health care is a high-risk industry (Helmreich & Davies, 2004; Kohn et al., 2000), by most accounts there is excessive process variation (Institute of Medicine, 2004; Kohn et al., 2000; Reason, 2000) and unacceptably large quantities of adverse events (Institute of Medicine, 2004; Kohn et al., 2000; Zhan & Miller, 2003). As such, the majority of hospitals operate as low reliability organizations.

Removing "blame and shame" from the equation may encourage health care professionals to embrace the system approach and participate in error reporting for the expressed purpose of system improvement. For example, one study found that 41% of physicians are discouraged from or not encouraged to report medical errors (Blendon et al., 2001). Shifting to a system philosophy diminishes the appropriateness of the current physician hierarchy emphasizing the assumption of personal responsibility and accountability for failures (Helmreich, 2000).

Understanding the properties of HROs, the system philosophy is the ideal approach to address error within the complex adaptive health care system. Weick and Sutcliffe (2001) describe five "mindful" organizational attributes summarizing the typical HRO system approach for safe and effective operations. Reliable organizations characteristically: (1) stay preoccupied with reducing failures; (2) stand reluctant to simplify errors; (3) maintain a heightened sensitivity to latent failures; (4) remain committed to learning from failures; and (5) defer decisions to experts at various levels. These elements are often cited and discussed by the IOM to call upon the health care industry to accept the system approach to error prevention and investigation (Institute of Medicine, 2003, 2004, 2007b; Kohn et al., 2000).

ANATOMY OF ERROR

With technological advances signaling the new millennium of health care, we must concede that extraordinary damage results from ordinary errors (Chiles, 2002). Unmistakably, medical errors are a significant reality (Kohn et al., 2000; Rasmussen, 1999; Reason, 2000) and a frequent product of poorly constructed health care delivery systems (Helmreich, 2000; Kohn et al., 2000; Reason et al., 2001). David Eddy, a nationally recognized physician patient safety expert, concisely summarized the impact modern demands have created with technological improvement and knowledge generation by stating, "the complexity of modern medicine exceeds the inherent limitations of the unaided human mind" (Millenson, 1997, p. 75).

In an evaluation of accidents within the context of a system, Reason and Hobbs (2003, p. 39) believe errors result from "the failure of planned actions to achieve their desired goal, where this occurs without some unforeseeable or chance interventions." Reason et al. (2001) describe this failure as a type of "vulnerable system syndrome" and emphasize early identification, evidence-based treatment, and prevention strategies to address process and system failures. To facilitate meaningful discussion, errors first need to be defined and accurately described (Benner et al., 2006; Wiegmann & Shappell, 2003). Broadly stated, there are two types of errors – latent and active (Reason, 1990, 2000).

Both latent and active errors work to disrupt systems damaging both patients and clinicians by contributing to adverse events and poor outcomes. As systems fail in a "step-by-step" fashion analogous to the cracking of metal under intense stress (Chiles, 2002), the evolution of a sentinel event is a system of contributory fractures. To explain this complex system of layers, fractures, and pressures, we offer the *Health Care Error Proliferation Model* illustrated in Fig. 1.

This model, incorporates several important elements of Reason's work, depicts the health care system segregated into defensive layers within the complex adaptive system as well as part of the global health care environment. Organization leaders are positioned at the blunt-end, while the clinician works closest to patient bedsides and resides at the sharp-end. The holes in each layer provide opportunity for error to manifest when health care professionals are unable to defend these system gaps at various organizational levels. Vigilant clinician error defenses are analogous to systematic survival techniques developed through experiential learning and reflection. Frequently, these defenses derive from localized workarounds, described as clinical improvisation (Hanley & Fenton, 2007), purposed to

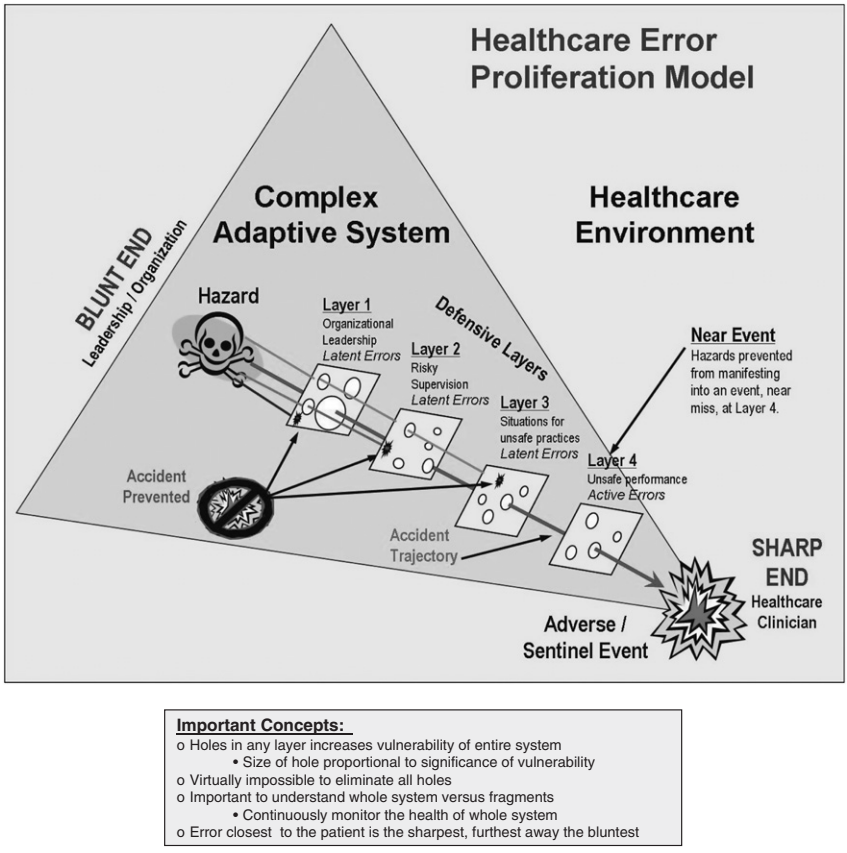

Fig. 1. Health Care Error Proliferation Model. *Source:* Based on Concepts from Reason (1990, 1998). Artistic and Graphic Assistance by Larry Reising.

repair gaps produced by actions, changes, and adjustments fashioned at higher defensive layers.

Latent Conditions and Errors

The least visible but most frequent type of error can be described as latent (Rasmussen, 1990, 1999; Reason, 1990, 2000). Latent errors can be defined as those less apparent failures of processes that are precursors to the

occurrence of errors or permit them to cause patient harm. Latent conditions exist distant from the delivery of patient care. For example, commonly observed latent conditions include operational glitches caused by poorly designed surgical time-out procedures, faults created by flawed patient identification policies, and inadequate resources allocation such as staffing, equipment, and supplies. Problematic or latent conditions may remain dormant for extended periods of time (Reason, 1990) as errors do not manifest themselves until the right circumstances arise (Reason, 2000; Wiegmann & Shappell, 2003).

Latent conditions have been characterized as situations placed in the system as part of its design or caused by actions taken by decision makers removed from the direct provision of care (Sasou & Reason, 1999). "Latent conditions are to technological organizations what resident pathogens are to the human body" (Reason, 1998, p. 10). Latent errors are literally "accidents waiting to happen" (Whittingham, 2004) in the absence of attention and treatment. As such, latent conditions are present as hidden malfunctions in organizational systems (Reason, 1990) contributing to the occurrence of adverse clinical events that are potentially harmful to patients (Rasmussen, 1990; Reason, 1998, 2000). Latent errors provide early and perhaps repetitive warnings of imminent accidents of consequence (Reason, 1998). Thus, in relation to health care systems, latent error identification and intercession provides an important adverse event prevention strategy (Cook & O'Connor, 2005; Cook & Woods, 1994; Reason et al., 2001).

Active Conditions and Errors

Active errors are actual breaches in the system defenses arising from the dormant latent conditions energized by some type of stimuli (Reason, 1990). Revealed through immediate feedback (Reason, 1998), active errors are usually related to clinical tasks manifesting out of patient care activities such as administering an intravenous antibiotic or performing a knee replacement surgery. Often, active errors result when latent strategic and operational decisions made at the highest organizational levels (Reason, 1998; Reason et al., 2001) shape a potentially dangerous setting for clinicians to chance. The active errors act like small holes in a water container. Occasionally, improvisation is utilized such that moving a single finger over a hole offers a temporary workaround and water stops leaking (Hanley & Fenton, 2007). Over time, however, more holes will likely materialize and soon the professionals can no longer mange the workaround. Although, the circumstance

necessitating a workaround can temporarily alleviate local symptoms from a systemic failure, this localized improvisation can worsen or potentate the overall decline in a complex system. As with other industries, health care professionals can either manage uexpected events ineffectually (Weick & Sutcliffe, 2001) as if covering a hole with a finger, leading to future accidents or health care professionals can pull the system into correction adverting future debacles.

Similar to other industries, sentinel-like events are virtually impossible to predict and even more difficult to scientifically study (Reason & Mycielska, 1982). Even though active errors appear more common due to the immediate result, latent errors are actually more prevalent (Reason, 1998) but less distinguishable without diligent surveillance activities (Lawton & Parker, 2002), such as robust occurrence reporting, failure mode effect analysis (FMEA), or root cause analysis (RCA). Hence, focused efforts to correct active errors provide little benefit to the system going forward (Rasmussen, 1990; Reason et al., 2001), especially from the future patient's perspective. While most health care organizations find surveillance activities difficult to master, those with a sensitivity to operations and a demonstrated concern for discovering the unexpected prove successful in receiving active reporting of discrepancies (Weick & Sutcliffe, 2001).

The practitioner's perspective for error and error causation makes a difference in the robustness of surveillance. Through providing practitioners with clear guidance for detecting, discussing, and reporting error, as well as frequently soliciting feedback, system issues begin to illuminate for easier identification. This perspective is an important aspect to identify early issues related to two situations which might arise when clinicians become aware of an error but may choose not report. The first situation is the "error of judgment and planning" when clinician performance progresses as intended but the overall plan is flawed leading to an error. The second situation is the "action not as planned" that describes poor clinical execution that occurs despite the presence of a good plan (Reason & Mycielska, 1982). When patient outcome is not impacted by either of these situations, clinicians may not report the error even though the chances of recurrence are significant.

There are several explanations suggested as to the reason why clinicians elect not to report near-errors or actual errors where patient harm was adverted. The clinician perception about what information should be reported is discussed in the literature. For example, when an active error causes injury practitioners generally believe it most appropriate to report "what happened" but not necessarily "how it happened" (Espin, Lingard, Baker, & Regehr, 2006). Surveillance activities that highlight active errors,

or the "what happened" have been favored to process analysis for the "how it happened" leading to forfeited improvement prospects. This active error emphasis perpetuates system volatility (Reason et al., 2001). Attention to correcting latent errors directly correlates to error reduction and systemic improvement (Institute of Medicine, 2004; Kohn et al., 2000; Reason, 1990, 2000) frequently attributed to reliable system processes (Weick & Sutcliffe, 2001).

The Proximity of Errors to the Adverse Event: Blunt and Sharp-Ends

Latent and active errors are described, in part, by the system element they arise from, as well as, by their proximity to an adverse event. As illustrated in Fig. 1, latent errors usually remain distal to the adverse event site (Reason, 2000; Reason et al., 2001). Within the complex health care system, distinct failure causation, and event prevention opportunities exist at varying levels of the organization (represented in Fig. 1 as the layers of defenses). There are four defensive layers in the model. As these layers are discussed in the following section, it is important to remember that each layer may contain multiple sublayers and other complex attributes not specifically discussed in this chapter.

At the first macroscopic layer (layer 1), leaders make organizational decisions about policy, procedure, and clinical function. These decisions potentially change, create, and/or eliminate holes at different levels within the complex adaptive system. Then, the second layer (layer 2) represents the supervisory and management role in the context of localized operations. These managers direct and organize the localized operations with varied strategies. Next, layer 3 represents the zone where policy, procedural, and environmental imbalances impact clinical practice. The resulting practices may contribute to the systematic error trajectory and interactions with system defenses. Finally the last defensive layer, layer 4, is the proverbial "rubber meeting the road" layer where unsafe clinical acts can result or lead to adverse events. At this microscopic level, clinicians work at the sharp-end while often not realizing they are protecting the patient from the unintended consequences created by multiple failsafe breakdowns.

The large triangle in Fig. 1 represents the complex adaptive health care system. Complex adaptive systems characteristically demonstrate self-organization as diverse agents interact spontaneously in nonlinear relationships (Anderson, Issel, & McDaniel, 2003; Cilliers, 1998), where professionals act as information processors (Cilliers, 1998; McDaniel &

Driebe, 2001) and co-evolve with the environment (Casti, 1997). Health care professionals function in the system as diverse actors within the complex environment utilizing different methods to process information (Coleman, 1999) and solve systemic problems within and across organizational layers (McDaniel & Driebe, 2001). Self-organization emerges as clinicians to adjust, revise, and rearrange their behavior and practice to manage the changing internal and external environmental system demands (Anderson et al., 2003; Cilliers, 1998) utilizing experiential knowledge (McDaniel & Driebe, 2001) and improvisation (Hanley & Fenton, 2007). Health care environments supporting and accepting self-organization produce better patient outcomes (Anderson et al., 2003). As such, care delivery system self-organization attributes, represented by the triangle in Fig. 1, impact the recognition, mitigation, and prevention of latent and active errors.

Next, the large square reflects the health care environment including the influences and forces outside the organization such as regulatory boards, consumers, payers, legislators, and others. The impact of payment systems, litigation trends, and evidence-based practice changes all contribute to external complexity that impacts individual health care organizations (Davis et al., 2002; Kohn et al., 2000). Although we describe some of these features within this chapter, the specific application of this external environment model attribute is better suited for another discussion.

A particularly valuable and integral aspect of our model is the attention to both the blunt-end and the sharp-end of error causation. Latent errors tend to reside closest to the triangle's blunt-end (left side) and represent the organizational level attributes. Errors manifesting at the clinician level develop as active errors or possibly near-events. A near-miss event is the nomenclature to describe those errors coming close to injuring a patient with harm adverted by a "last minute" defensive action close to the bedside.

At the other end of the triangle and adjacent to the accident, active errors intimately link to an adverse event at or closest to the patient (Cook & Woods, 1994). The apex or pointed-end of the triangle resides closest to the accident. If the triangle were metal, the pointed-end would be sharp; hence, the term "sharp-end" is used to describe active errors. Also, the term sharp-end error is metaphorically analogous to the blame and the punishment frequently exacted on the health care professionals for human error (Kohn et al., 2000; Reason, 2000). The specific relationship of the sharp-end and the blunt-end concepts to adverse events and the defensive layers will be further developed in the sections to follow. Active and latent errors are impacted and influenced, directly and indirectly, by the internal defensive layers as well as the entire complex adaptive health care system.

HEALTH CARE ERROR PROLIFERATION MODEL

In this section, we adapt the *Swiss Cheese Model* (Reason, 1990) to health care organizations in which we call the *Health Care Error Proliferation Model* (Fig. 1). The *Swiss Cheese Model*, likens the complex adaptive system to multiple hole infested slices of Swiss cheese positioned side-by-side (Reason, 1990, 2000). The cheese slices are dubbed defensive layers to describe their role and function as the system location outfitted with features capable of intercepting and deflecting hazards. The layers represent discrete locations or organizational levels potentially populated with errors permitting error progression. The four layers include: (1) organizational leadership, (2) risky supervision, (3) situations for unsafe practices, and (4) unsafe performance.

The *Swiss Cheese Model* portrays hospitals as having multiple operational defensive layers outfitted with essential elements necessary to maintain key defensive barricades (Cook & O'Connor, 2005; Reason, 2000). By examining the defensive layers attributes, prospective locales of failure, the etiology of accidents might be revealed (Leape et al., 1995). Experts have discussed the importance of examining these layers within the context of the complex adaptive health care system (Kohn et al., 2000; Wiegmann & Shappell, 2003) as illustrated in our *Health Care Error Proliferation Model*. The contribution that the complex system provides to error was suggested in a 2001 survey in which 76% of nurses indicated that their inability to deliver safe health care was attributed to impediments created by unsafe working conditions (American Nurses Association, 2006). This data reflects the general presence of unsafe working conditions in health care facilities across the nation. There is probably an operational disconnect at many organizational levels and between multiple disciplines created by the expansive system complexity while under intense financial, human, and political pressures.

The holes in the cheese represent actual areas where potential breakdowns or gaps in the defensive layers permit hazardous error progression through the system (Cook et al., 2000; Reason, 1990). These holes continuously change size and shift locations (Reason, 2000). When gaps at each layer sequentially align, the system fails. In other words, sentinel events occur when multiple individual, yet small faults, come togather to create the circumstances for a hole alignment sufficient to produce total system failure (Cook & O'Connor, 2005). This unobstructed hole alignment occurs infrequently. In the vast majority of situations, health care clinicians control these holes and defend the system integrity (Cook et al., 2000) to

stop error progression. Only when the practitioner is unable to anticipate, detect, and impede hazards from passing through the holes will an adverse event manifest (Reason, 1998). However, given the substantial number of activities involved in delivering patient care (Institute of Medicine, 2004), coupled with the heavy workloads of clinicians (Joint Commission on the Accreditation of Healthcare Organizations, 2007), individual holes or gaps within specific layers represent the normal consequence of complexity and operational demands. The alignment of the holes to create an adverse or sentinel event is analogous to the concept of the "perfect storm."

Using the concepts illustrated in Fig. 1, we discuss in the following sections the organization, leader, manager, and clinician contribution to strengthening and protecting the defensive layers along the continuum from the blunt-end to the sharp-end of adverse events. When an accident manifests, the presence of latent factors may be revealed through careful and unbiased examination of the system (Farrokh, Vang, & Laskey, 2007) by methodically evaluating each defensive layer (Reason, 2000). Hazards blocked within or at any one of the first three defensive layers is termed a latent error (Reason, 2000), while failure blocked at the fourth layer, virtually at the patient's bedside, is termed a near-event (Cook et al., 2000; Cook & Woods, 1994; Reason, 2000). The culmination of error proliferation is heavily dependent not only on the layers of interaction but interdependent to the general culture of both the health care system internally, at the organization level, and externally, at the professional, legal, and delivery system levels.

Layer 1: Organizational Leadership

The first layer, most distant from the adverse event, is leadership at the highest organizational level. This level of defense is the most challenging to alter in health care organizations (Nolan, Resar, Haraden, & Griffin, 2004) because the current "blame and shame" person approach is frequently derived from an attribution like process (Reason, 1998; Sasou & Reason, 1999). Leaders often ascribe responsibility for faulty processes or adverse events as resulting from the lack of effort (Reason, 2000), inability (Vincent, 2003), incompetence (Rasmussen, 1990, 1999), and absence of vigilance (Reason & Hobbs, 2003) to the clinical professional. As such, attribution for poor performance, or "practice breakdowns" results in actions aimed at clinicians instead of the decisions produced by leaders.

Effective leadership is necessary to maintain safe systems (Joint Commission Resources, 2007). Leaders are accountable for "engineering a

just culture" as a critical aspect to providing safe patient care (Reason, 2000). The Institute of Medicine (2004) speaks to three vital aspects of a leader's responsibility to shape their institutional culture of safety. First, leaders need to recognize and accept that the majority of errors are created by the system that they cultivate and direct (Institute of Medicine, 2004; Reason, 2000). Second, the role of leadership should include the perceptible daily support for practitioners (Institute of Medicine, 2004; Weick & Sutcliffe, 2001). Third, leaders should sincerely embrace and inculcate continuous organizational learning (Hofmann & Stetzer, 1998; Institute of Medicine, 2004). In combination, these three IOM recommended qualities not only create efficient organizations but they also help reduce latent and active errors.

Accordingly, the Institute of Medicine (2004) suggests several focus areas for strengthening the organization. These measures include: adopting evidence-based and transformational leadership practices, increasing workforce capability, enhancing workplace design to be error resistant, and creating a sustainable safety culture as necessary elements for vigorous safety defenses. Developing a robust error reporting system or incorporating prospective process reviews by quality teams are just two examples of leaders positively influencing organizational acceptance of a patient safety culture.

In order to better understand the behavior of individuals, leaders must attempt to determine what subordinates are thinking about situations (Pfeffer, 1977). Green and Mitchell (1979) developed a model to study attributional leader traits. Leader behavior is a consequence of interpretation relevant to subordinate performance. In attempting to understand how subordinate performance affects leader reaction, it is vital to determine the cause of subordinate performance whether good or poor. There are four causes that may or may not be within the subordinate's control: competence, effort, chance, and other uncontrollable external causes. Both competence and effort are internal causes of performance, while chance and other uncontrollable causes are outside the subordinate's control (Green & Mitchell, 1979).

Causality is attributed more to the subordinate than to the situation if the subordinate has had a history of poor performance. Additionally, if the effects of the poor performance have severe outcomes, it could be determined that the subordinate was at fault (Mitchell & Wood, 1979). In this situation, the leader would focus more on some remedial action towards the subordinate, rather than on the situation (Bass, 1990). Continued punishment may lead to a downward spiral in performance leading to learned helplessness which would be dealt with even more punitively if a

subordinate was previously viewed as a poor performer (James & White, 1983). Unfortunately, this punitive punishment can lead to even greater declines in subordinate performance (Peterson, 1985) and possibly greater system instability.

Developing and supporting a culture of excellence begins at the leadership level by embracing a preoccupation with preventing failure (Weick & Sutcliffe, 2001). Chiles (2002, p. 15) describes a leader's preoccupation with system cracks as the "chess master spend[ing] more time thinking about the board from his opponent's perspective than he does from his own." Leaders ought to seek assistance from clinicians and other health care professionals with expertise in detecting and extracting the subtle signals of impending issues from the constant noise of routine workflow prior to initiating systematic change. These experts can provide the feedback necessary to proactively identify and correct system abnormalities with minimal disturbance given that the best intended system changes can be quite disruptive to the institution.

Layer 2: Risky Supervision

The second defensive layer is management, specifically those supervisors at various organizational levels directly responsible to the leadership team for system metrics and outcomes. When management is ineffective in correcting known gaps or holes, patient welfare is endangered. Thus, management involvement is critical in maintaining a defense against adverse events (Reason, 1990). Risky supervision arises when management decisions lead to circumventing or not enforcing written policies and procedures (Reason & Hobbs, 2003), overlooking known issues, and the absence of corrective actions (Leduc, Rash, & Manning, 2005) as employees engage in potentially consequential unsafe activity. Practitioner deviations from written policies and procedures significantly contribute to adverse events (Vaughn, 1996). As such, management's enforcement of policies is critical to the safe functioning of any organization.

Workload pressures created by understaffing or overly stressful work assignments can lead to poor outcomes (Aiken et al., 2001; Aiken, Clarke, Sloane, Sochalski, & Silber, 2002; Stone et al., 2007). Performance pressures caused by any number of issues creating a difficult and error-prone workplace are examples of risky supervision. In 2002, the Joint Commission reported the lack of nursing staff contributed to nearly 25% of the unanticipated issues resulting in patient death or injury (Joint Commission on Accreditation of

Healthcare Organizations, 2007). Tucker and Spear (2006) reported inadequate nursing task times, multiple unplanned changes in task, and frequent interruptions mid-task throughout a typical shift. These reported issues may result in harm to patients, especially in complex care situations such as the intensive care or perioperative environments. In addition, unpaid overtime at the end of the shift is a well known phenomenon (Tucker & Spear, 2006) with nurses reporting on average greater than one hour per shift (Rogers, Wang, Scott, Aiken, & Dinges, 2004; Tucker & Spear, 2006). The intensity of work coupled with the inability of nurses to complete their work within a scheduled shift indicates poor job design and inadequate support, possibly at the level of immediate supervision. Also, these work pressures are not unique to nursing since other health care professional face similar conditions (Institute of Medicine, 2004; Kohn et al., 2000).

In short, managers, both clinical and administrative, are directly responsible for organizing, implementing, and evaluating policies and procedures to guide the safe supervision and management of the complex adaptive care delivery system. Considering system design, establishment of boundaries, and expectations under which clinicians practice, management can foster safer performance through more supportive organizational norms and values (Institute of Medicine, 2004; Kohn et al., 2000; Rasmussen, 1999; Reason et al., 2001). Supervisors should foster a safe environment by actively identifying and addressing deficiencies among practitioners, equipment, processes, and/or training. Safe practices include correcting issues, utilizing progressive discipline and on-the-spot performance coaching immediately addressing unacceptable clinical behavior while promoting adherence to policies and procedures.

When considering the situations related to unsafe practices, supervisory level staff are integral to maintaining the sensitivity to the operations and commitment to the resiliency of the organization (Weick & Sutcliffe, 2001). Imperfect environmental conditions represented in the next layer, preconditions for unsafe practices can be revealed by vigilant supervisory and management participation. Thus, situational awareness of their respective environment in relation to the "big picture" is vital to managers' impact on the next layer.

Layer 3: Situations for Unsafe Practices

The third layer represents precursors to practices that are not safe. Careful attention to error substance at the situations for unsafe acts level is

necessary to protect the system and the clinicians from mistaken error attribution. Substandard conditions and practices are two situational circumstances responsible for the presence of unsafe acts in practice (Reason, 1998). In fact, these two synergistic conditions each fuel the other as situations increasing the likelihood of an unsafe act (Reason, 1990). The inability of local systems to provide practitioners with reliable support, including information, equipment, and supplies, has been described as "operational failures" (Tucker, 2004) which create substandard conditions for practitioners attempting to execute their patient care responsibilities. Tucker and Spear (2006) observed that nurses experience an average of 8.4 operational failures per shift.

An example of a serious, but often stealth, operational failure situation is the "down-time" associated with an automated pharmaceutical dispensing system interface that links the dispensing machine to the pharmacy. When this operational failure occurs, the dispensing machine is unable to profile each patient's personal medication schedule. When a patient specific medication profile is absent a notable defensive hole is created. This issue increases the probability for error as the dispensing machine's entire pharmaceutical contents are readily available and the selection is not limited to only those medications specifically ordered for each patient. Notably, a clinician is able to remove the wrong medication or the wrong medication dose from the dispensing unit, an act that is less likely when the pharmacy-machine interface is properly functioning. As such, this situation creates the precondition for a potentially serious event as delivering the wrong medication or dosage in this example results in the occurrence of an unsafe practice that may lead to patient harm or death.

Other failures, or errors, at the clinician level result from difficulties in task partitioning exemplified by postponing a task until missing supplies arrive due to operational failures (Tucker & Spear, 2006) acting as a precondition to actualize clinician error. With task partitioning, clinicians experience work interruptions (such as not being able to administer an urgent or stat medication) linked to an unsafe precondition (reoccurring delays in pharmacy providing stat medications to the clinical unit) thereby leading to substandard practice (omission of medication delivery, possibly due to memory, or deterioration in patient condition, due to time). These examples illustrate the complexity of error at the clinician level that might be considered substandard practice or even a "practice breakdowns" (Benner et al., 2006). System focused organizations highlight the reluctance to accept simplifications like "practice breakdowns" in the pursuit of error causation. Informed organizations use strategies such as tracing

preconditions through the complex adaptive system, layer by layer, prior to the clinical practice level, seeking to determine why the event manifested versus rather than who was responsible (Weick & Sutcliffe, 2001).

In a concerted effort to universally improve the situations for safer clinical practices, many agencies and organizations advocated best practices and suggested system attributes purposed at addressing errors related to substandard practices. The Joint Commission, through the creation of the National Patient Safety Goals (NPSG), and the Institute for Safe Medication Practices (ISMP), through medication delivery recommendations have positively impacted health care in emphasizing the importance of patient safety. For example, sound-alike, look-alike drugs (SALAD) and high-alert medications have been identified as areas prone to compromise patient safety, especially when clinician vigilance is altered by other complex adaptive attributes, such as fatigue and workload. Strategies have been adopted to address issues related to substandard medication practices caused by unsafe situations. Recommendations can quickly become accreditation standards such as removing dangerous highly concentration drugs like potassium chloride from patient care units, adopting formal patient identification requirements, and mandating surgical instrument counts as an essential step at the end of surgical procedures. On the horizon, situations related to using automated pharmaceutical dispensing devices, incorporating bar-scanning technologies into the medication delivery process, and implementing electronic medical records all promise to reduce opportunities for unsafe practice.

Layer 4: Unsafe Performance

The fourth layer represents unsafe practices which are usually linked to the health care professional's performance at the "sharp-end" of care leading to an adverse event. In the simplest form, unsafe practices consist of errors and violations (Reason, 2000). Errors frequently involve perceptual (Wiegmann & Shappell, 2001), decision making (Norman, 1988; Perrow, 1984; Rasmussen, 1990), and/or skill-based lapses (Norman, 1988; Reason, 1990). Unsafe practices are the tip of the adverse event iceberg (Reason, 1998) as the number of these acts tends to be rather large but discrete (Reason, 1990, 1998) and usually are the most noticeable following a sentinel event (Perrow, 1984; Reason, 1998, 2000; Wiegmann & Shappell, 2003). Although the error and sentinel event numbers have yet to be

accurately quantified, the magnitude of unsafe practices is put into perspective by considering the IOM's (2007b) finding that on average, each hospitalized patient experiences one medication error per day.

Violations are errors that are more serious in nature but occur less frequently (Reason, 1990). Wiegmann and Shappell (1997) describe violations as the willful disregard of policies, procedures, and rules governing health care professionals at both the organizational and regulatory levels. Violations are willful, therefore, missing the sincere intention to enhance a problematic situation through improvisation. Reason (1998) attributes violations to issues related to motivation, attitude, and culture. These acts amount to deviations from safe operating practices (Vincent, Taylor-Adams, & Stanhope, 1998) even possibly resulting from a lack of training or education (Institute of Medicine, 2004). Examples of clearly willful clinician violations amounting to "practice breakdowns," are purposely falsifying documents, failing to execute life-saving procedures when observably necessary, and stating untruthfully that actions, events, or treatments were completed.

At this layer, clinicians develop capabilities to detect, contain, and report unsafe situations or errors. Expert clinicians maintain their distance from the sharp-end of an adverse event with heightened situational improvisation (Hanley & Fenton, 2007) to generate a protective barrier despite numerous complex care management issues (Kohn et al., 2000; Vincent, 2003). This protection essentially results from experience and heightened familiarity with improvisation to mitigate issues created by lack of clear protocols for a situation, high stress clinical scenarios, and fatigue from heavy workloads and long working hours (Hanley & Fenton, 2007; Helmreich & Davies, 2004; Vincent, 2003). Left to novice clinician hands, improvisation may lead to unfavorable consequences, as important experiential knowledge has not developed.

It is suggested that error is the symptom of a practice breakdown (Benner et al., 2006). The complexity embedded within the situation when an error is unsuccessfully defended by an expert clinician at the sharp-end is substantial. Specifically, incorporating improvisation as a normal practice attribute might render the system better capable of defending from error. Health care professionals should be expected to draw upon their prior experiences (Weick, 1993) in order to positively impact troublesome situations with the resources at hand (Hanley & Fenton, 2007) even if it requires them to not follow policies and procedures. When system opportunity for failure exceeds the clinician's ability to workaround and minimize error, the risk for a system meltdown is substantial (Chiles, 2002;

Kohn et al., 2000). Expert health care professionals when found at or near the sharp-end of a sentinel event signals the likely presence of a system overload rendering ineffective the expert's improvisation skills.

Weick and Sutcliffe suggest that organizations embrace "the reluctance to accept error simplifications responsible for creating more complete and nuanced pictures" as an important HRO characteristic (2001, p. 11). There are times when occurrences considered to be violations are actually issues related to motivation, attitude, and culture (Reason, 1998). Some acts amount to deviations from safe operating practices (Vincent et al., 1998) while others probably result from a lack of training or education (Institute of Medicine, 2004). Some might argue leaders and managers responsible for safe staffing and correcting hazards are responsible for the commission of a violation when patients are harmed due to known but uncorrected issues (Reason, 1990, 1998).

Willful practice violations assume the presence of an informed and trained professional with an unobstructed understanding of the policies and procedures (Amalberti, Vincenct, Auroy, & de Saint Maurice, 2006). The professional must be working in an environment where the specific policy and/or procedure is enforced, and the professional knowingly and deliberately disregarded the rule or rules (Reason, 1998). However, health care education and training specific to patient safety is lacking in many curriculums (Institute of Medicine, 2003) and it is an important aspect to view adverse events in the context of error versus willful violation. As such, the significant issue facing organizations, even society in general, is they do not recognize errors as an unavoidable aspect of limitations in human performance (Reason et al., 2001). Despite the negative impact upon the complex adaptive system when human error manifests, the overall impact of excellent patient care produced by well-intended clinicians should outweigh the consequence caused by unintentional error.

Defending the Patient from System Errors

Although problem solving and short-term fixes correct errors that arise in the normal course of task completion (Tucker & Edmondson, 2003), these fixes are not typically reported as clinicians improvise and adapt in the normal course of work. Consider the situation created when important medications are unavailable to nurses, yet the patient's condition requires availability. The result is a clear gap in the medication delivery system.

By improvising, the clinician could borrow the same medication from a different patient's medication bin (which could be considered a violation of policy). Time permitting, the nurse could leave the busy unit in order to walk to the pharmacy. This fills the existing gap, or need, but may create yet another if another patient deteriorates while the nurse is off of the unit. It is important to recognize that eliminating active failures stemming from unsafe acts prevents only one adverse event (Reason, 2000) and does not construct better defenses for removing system gaps (Cook et al., 2000). Therefore, examining and adjusting the system at the organizational and managerial layers to "manage error" is a more effective method for eliminating a larger variety of latent errors (Amalberti et al., 2006), subsequently limiting the manifestation of active errors (Reason, 1998).

To prevent adverse events that result directly from unsafe acts requires blocking holes or trapping hazards that result from within the clinician's controlled defensive layer. The most effective method to mitigate any error-prone process is to create and support an occurrence or incident reporting system (Cook & O'Connor, 2005; Department of Veterans Affairs, 2002). Sasou and Reason (1999, p. 3) state, "failures to detect are more frequent than failures to indicate and correct." However, leaders must support the removal of blame and punishment if error reporting and detection is to become routine in practice (Hofmann & Stetzer, 1998; Kohn et al., 2000). In an effort to protect the system and the patient from error, it is perhaps even more important to review the organizational process when near-events occur, those situations where an adverse event is avoided at the very last moment. Contrary to contemporary patient safety knowledge, professional and quasi-legal bodies appear less concerned with the category of error resulting in the near-events or nearly harmful to patients instead favoring those situations resulting in actual patient injury. This focus might be explained by the regulatory body predisposition to allow the consequences of an accident to "color and even distort" the perception of the system attributes and the actions of the clinicians leading up to the event (Reason & Mycielska, 1982). These almost accidents, or near-events, present significant evidence of an approaching sentinel event like the tremor felt prior to an impending earthquake. Reason (2000) considers near misses as free lessons that previews or represents potential and preventable future adverse events. Organizations must be alert as well as vigilant in identifying these near misses in order to correct, change, or modify policies, procedures, and practices to prevent future errors that could result in a sentinel event.

THE ROLE OF HEALTH CARE LEADERSHIP IN PATIENT SAFETY

Since its formalization in 2000, the patient safety movement gained both national attention and increased resources dedicated to creating safer health care delivery systems. In 2003, the Joint Commission established the initial NPSG. These initiatives helped patient safety advocates to actively transform health care based on an understanding from the organizational accident and human factors sciences. This transdisciplinary knowledge transfer assists leaders in understanding the path to develop health care into a high reliability or high risk but low accident industry (Institute of Medicine, 2003; Kohn et al., 2000). Developing and maintaining the optimal patient safety culture while concurrently managing the overall organization performance metrics is challenging (Joyce, Boaden, & Esmail, 2005; Reason & Hobbs, 2003). In a 2004 survey, hospital administrators ranked financial challenges at the top of the list of their 11 priorities while patient safety ranked 10th and patient satisfaction ranked 11th (American College of Healthcare Executives, 2005).

Over the last decade, health care organizations attempted to eliminate error through the zero-deficit culture (Institute of Medicine, 2004) with modest success (Nolan et al., 2004; Roberts, 2002). As a direct result of the flawed human approach to error management, health care professionals hide errors in order to protect themselves from reprisal and punishment (Kohn et al., 2000). Unreported latent errors exist partly due to the organizational inertia (Palmieri, Godkin, & Green, 2007) created by fear (Kohn et al., 2000; Millenson, 2003), and remain free in the environment until a system meltdown results in an adverse event (Reason, 1998; Reason et al., 2001; Vaughn, 1996, 1999).

While the NCSBN seeks to establish the attentiveness and surveillance standard of nursing practice associated with watchful vigilance in preventing the failure to rescue (Benner et al., 2006), health care leaders are familiar with clinician inability to effectively monitor patients within the complex adaptive system due to a variety of pressures. The increased complexity of the typical hospital unit coupled with reduced nursing hours due to financial, human resource, and simple supply constraints and increased physician workload is problematic to effective and safe hospital operations. As a result, health care leaders should recognize that patient safety experts are unable to uniformly define a realistic "one case fits all" threshold for system error versus professional neglect in the contemporary health care system.

Professional and governing organizations for pharmacy, medicine, and nursing play pivotal roles in providing support to all constituents in advocating the system approach versus their traditional person approach to event investigation and adjudication. Health care leaders are well positioned to support clinical professionals and improve patient safety by vigorously advocating that error causation investigations related to adverse events embrace the system approach. Some organizations remain attached to the human approach as evidenced by statements such as, "when nursing practice falls below this minimal acceptable threshold sanctions are necessary to protect the public" (Benner et al., 2006, p. 59). The differentiation between willful and accidental error is an essential point for consideration in any RCA investigation. Learning about near-events and latent errors necessitates an open, honest, and sincere organizational culture.

The Institute of Medicine (2004, p. 226) states, "although near-miss events are much more common than adverse events – as much as 7 to 100 times more frequent – reporting systems for such events are much less common." In order to turn the potential lemon of near misses in the lemonade of error prevention, leaders should consider defining what a near miss is in their organization, talk openly about near-events when they occur, and preach near-events are positive signs of a system with functioning safeguards despite vulnerability (Weick & Sutcliffe, 2001). Preoccupied with probing their potential and actual failures related to nearly adverting disaster (Weick & Sutcliffe, 2001), HROs realize these nearly missed events create future success for all parties.

Leaders should also recognize that the vast majority of patient safety knowledge, including the NPSGs, are based on expert opinions, recommendations, and experiences as well as anecdotal knowledge (Agency for Healthcare Research and Quality, 2001). The Agency for Healthcare Research and Quality (AHRQ) analyzed the current state of affairs for patient safety practices of which 79 practices were reviewed for the strength of research evidence supporting current recommendations. After review, only 11 patient safety practices were correlated with a high level of reliable evidence. These 11 practices were clinical in nature and did not include crew resource management, computerized physician order entry, simulation use, or bar-coding technology. In addition, AHRQ found the literature is insufficient to make rational decisions about organizational nursing environments. A number of recommended practices with longstanding success outside of health care were not included in the analysis because of the general weakness of health care evidence. These practices included incident reporting, application of human factors principles, utilization of

communication models, and promoting cultures of safety. Further research, especially for practices drawn from industries outside of health care, is needed to fill "the substantial gaps in the evidentiary base" of patient safety practices (Agency for Healthcare Research and Quality, 2001).

Health Care Leaders and High Reliability Organizations

Through "problemistic search" (Cyert & March, 1963, p. 120) managers and leaders scan the environment for quick solutions to immediate problems regardless of complexity. Reason (2000, p. 770) states, "most managers … attribute human unreliability to unwanted variability and strive to eliminate it." Effective error management includes decreasing the incident of critical error (Reason, 2000) and developing improved systems capable of containing dangerous error (Reason et al., 2001). When management demonstrates positive organizational safety practices, subordinates' will accept safety as an essential job responsibility considerably improves (Dunbar, 1975; Katz-Navon, Naveh, & Stern, 2005; Kohn et al., 2000). This approach, reflective of HROs, focuses attention to learning and awareness at all levels of the organization (Reason, 1997), including the individual, department, and leadership. Preoccupied with probing their potential and actual failures related to nearly adverting disaster, HROs frequently avoid sentinel events (Weick & Sutcliffe, 2001) as these nearly missed events create future successes from a single realization. In fact, an organization should develop a policy specifically endorsing the system philosophy to error with emphasis on the non-punitive consequences related to unintentional errors (Helmreich & Davies, 2004).

HROs exhibit resilient systems and superior hazard intelligence (Reason, 2000) where practitioners employ positive adaptive behaviors (Mallak, 1998) to immediately defend the patient from potentially harmful situations therefore sustaining the reliability of the process (Carroll & Rudolph, 2006). Linking the actual error to organizational factors is difficult (Wiegmann & Shappell, 2001) as system failure results from the misguided dynamics and imperfect interactions of the care delivery system (Institute of Medicine, 2004). Recognizing the importance of developing safe and reliable health care delivery systems (Institute of Medicine, 2004) necessitates active involvement by multiple health care stakeholders such as practitioners, administrators, regulators, and even patients (Katz-Navon et al., 2005; Leonard, Frankel, & Simmonds, 2004). The reliability of an organization is a measurement of process capability (Reason et al., 2001) within a complex

system (Cook & O'Connor, 2005) under the stresses of ordinary and unordinary conditions (Berwick & Nolan, 2006).

An elevated level of organizational vigilance is reflective of HROs as they generally support a cultural predisposition that prevents the materialization of adverse events (Weick & Sutcliffe, 2001). Hence, the investigatory work for organizations embracing the system approach deemphasizes active errors in order to scrutinize system malfunctions (Reason, 1990) in the quest to discover latent conditions and rejuvenate processes (Rapala & Kerfoot, 2005). Rigid hierarchies create a vulnerability to error and leaders should first seek to elimination error at their own defensive layer. In doing so, leaders prevent the culmination with lower level errors by deferring decisions to those with the expertise (skills, knowledge, and experience) consequently improving the overall system (Kohn et al., 2000; Reason, 2000; Weick & Sutcliffe, 2001).

Leaders focused on eliminating the majority of common error from their respective organization should recognize and understand the elements of Reason's work and our subsequent adoption to the Health Care Error Proliferation Model. Low incidences of accidents are observed and reported in HROs, most commonly found in aerospace (Nolan et al., 2004) and nuclear generation industries (Cook, Woods, & Miller, 1998). HROs experience fewer accidents then other organizations (Reason, 2000) as HROs recognize the importance of limiting human variability in their pursuit of safety (Roberts, 2002). In addition, HROs remain vigilant in recognizing the possibility of failures (Reason, 2000; Weick & Sutcliffe, 2001) and discover solutions in the work of human factor and organizational accident experts (Helmreich & Davies, 2004; Kohn et al., 2000). The increased diligence in examining systems for error is important. In recent literature, the concept of technological iatrogenesis (Palmieri & Peterson, 2008) and the resulting e-iatrogenic error (Campbell, Sittig, Ash, Guappone, & Dykstra, 2007; Weiner, Kfuri, Chan, & Fowles, 2007) has been suggested to be a significant future contributor to health care adverse events. As the health care technology enterprise rapidly expands, the likelihood of new and previously unseen error typologies impacting systems of care is probable (Harrison, Koppel, & Bar-Lev, 2007; Palmieri, Peterson, & Ford, 2007). This new error formation presents significant challenges for the system as Lyons et al. (2005) found that administrators, physicians, and nurses hold different beliefs about the universal assistance and the specific obstacles relevant to the overall information system as a clinical benefit.

The most important distinguishing feature of HROs is their collective preoccupation with identifying and eliminating system failure (Reason,

2000; Weick & Sutcliffe, 2001). The notion that HRO's are preoccupied with proactively identifying system flaws as an effective methodology for organizational improvement is not universally accepted, however. Some health care experts suggest "a systems approach is based upon a post hoc analysis and redesign of a system based upon unsafe performance" (Benner et al., 2006, p. 51). As the proactive FEMA increases in popularity with continued organizational interest in developing HRO attributes, the system approach to preventing failures will be universally recognized. Health care leaders remain ideally positioned to make the system approach a reality by transforming their institutions into HROs.

CONCLUSION

The *Health Care Error Proliferation Model* is based on the premise that health care organizations struggle to effectively integrate existing research, novel approaches, practice innovations, and new knowledge about patient safety into their complex care delivery systems (Gray, 2001; Kohn et al., 2000; Nolan et al., 2004). Health care clinicians and leaders are beginning to recognize the probable creation of more serious problems should the prevailing "punitive cultures and a focus on 'bad apples' instead of poorly constructed systems" be permitted to continue to impedes progress (Devers et al., 2004, p. 114). We reviewed a diverse body of knowledge and provided a transdisciplinary analysis of health care errors and events. Then, we discussed the relevant literature pertaining to the conceptualization of errors, approaches to identifying the causes of errors, and the utility of the Swiss Cheese Model of adverse events in reconsidering the existing health care leadership approach for developing organizational cultures of safety. Finally, we discussed the critical role of health care leadership in building defenses against adverse events by demonstrating the attributes of HROs.

We conclude that within the discipline of health care leadership, the value of learning and applying basic organizational accident principles (Leape et al., 1998) to the complex adaptive system of health care organizations is paramount to preserving patient safety (Helmreich & Davies, 2004; Reason, 2000). Embracing the system philosophy to error management is vitally important for health care leaders desiring to achieve HRO status. Health care professionals should support the patient safety movement by focusing on systems versus people as the fundamental strategy for developing into a high reliability industry.

We agree with Bennis (1989) who argued that organizational cultures are created, supported, and sustained through leadership role modeling of acceptable organizational conduct. Leaders create and sustain organizational safety by: (1) role modeling the acceptable behaviors; (2) embracing the system approach to managing critical organizational incidents; (3) rewarding desired and productive behaviors; and (4) eliminating the use of blame, shame, and punishment to address professional error. Understanding the Health Care Error Proliferation Model as a strategic overview for identifying latent and active error will improve the complex health care delivery systems. The responsibility for organization improvement rests in the capable hands of health care leaders to promote discovery, motivate and advocate transformation, and protect the integrity of systems and people in the pursuit of patient safety. As leaders continue to emphasize support for the system approach and push to reduce the person "blame and shame" approach, errors will decline. As errors decline, hospitals will continue their progress to demonstrate the characteristics associated with HROs. In the future the incredible number of patients who die each year due to health care errors will decline and the entire system will successfully improve the quality and safety of patient care.

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